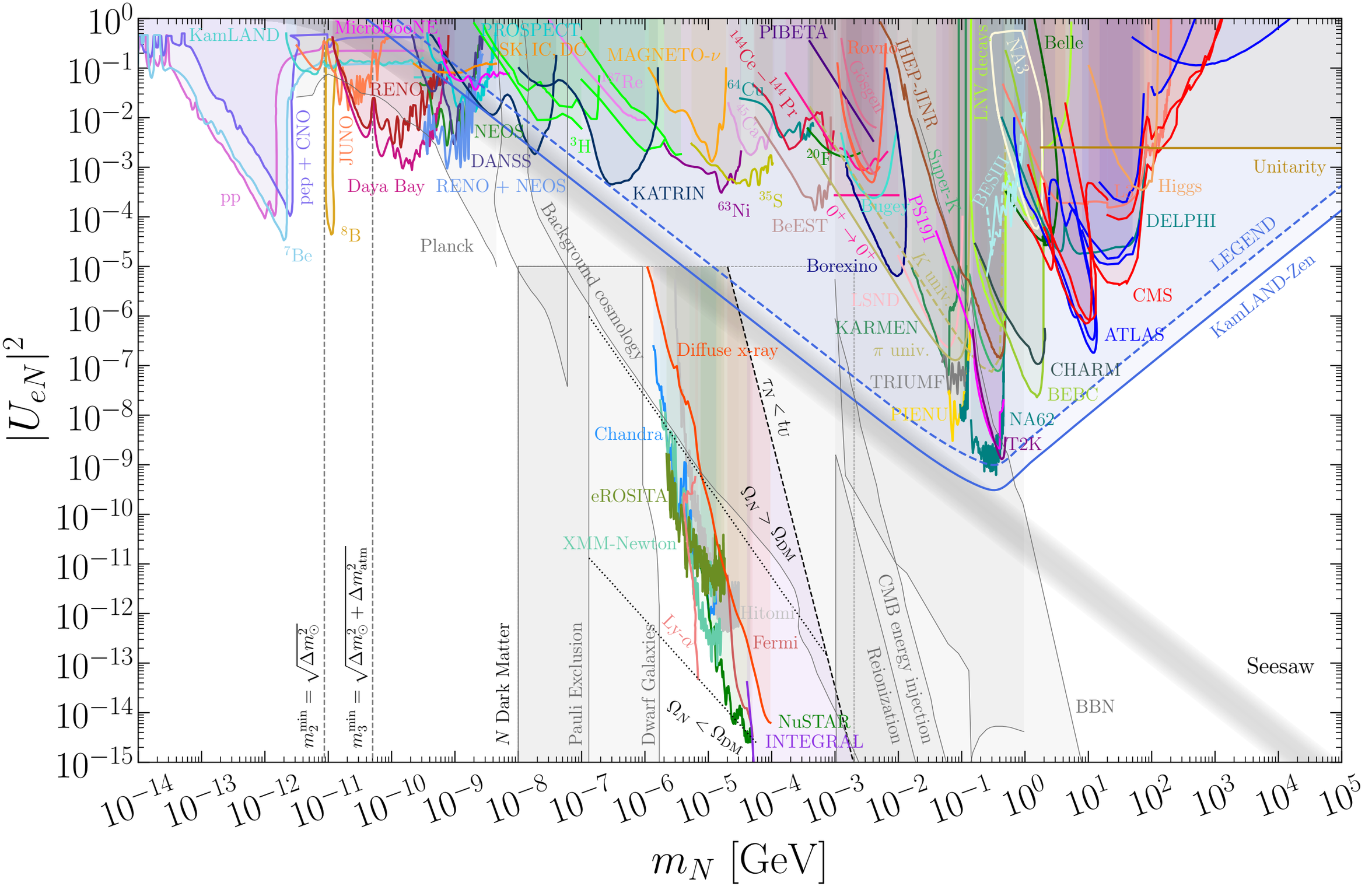

$|U_{eN}|^2$
m_N [GeV]
KamLAND
pp
^{7}Be
pep + CNO
MicroBooNE
JUNO
^{8}B
RENO
Daya Bay
PROSPECT
SK, IC, DC
NEOS
RENO + NEOS
DANSS
^{3}H
MAGNETO-ν
^{37}Ar
^{144}Ce – ^{144}Pr
^{64}Cu
^{45}Ca
KATRIN
^{63}Ni
^{35}S
PIBETA
Rovno
Gösgen
^{20}F
Bugey
BeEST
$0^+ \to 0^+$
IHEP-JINR
Super-K
PS191
K univ.
LNV decays
NA3
BESIII
Belle
Unitarity
Higgs
DELPHI
LEP
CMS
ATLAS
LEGEND
KamLAND-Zen
Planck
Background cosmology
Borexino
LSND
KARMEN
π univ.
TRIUMF
PIENU
CHARM
BEBC
NA62
T2K
Diffuse x-ray
$\tau_N < t_U$
Chandra
eROSITA
XMM-Newton
$\Omega_N > \Omega_{\rm DM}$
Hitomi
Fermi
Ly-α
$\Omega_N < \Omega_{\rm DM}$
NuSTAR
INTEGRAL
CMB energy injection
Reionization
BBN
Seesaw
N Dark Matter
Pauli Exclusion
Dwarf Galaxies
$m_2^{\rm min} = \sqrt{\Delta m^2_\odot}$
$m_3^{\rm min} = \sqrt{\Delta m^2_\odot + \Delta m^2_{\rm atm}}$